🎯 ObjectivesHow to Speak:

“Our objective is to build a machine-learning model that can detect depression using voice features like tone, pitch, and pauses.
Initially, we’ll train the model on English data, and later expand it to include Indian regional languages like Kannada.
We’ll also integrate facial emotion features to check if combining voice and facial expressions improves accuracy.
Finally, our aim is to design a simple early-screening tool that can be used in clinical or educational setups for fast and non-invasive depression detection.”

🌍 Scope & Feasibility How to Speak:

“The scope of our project is to make depression detection accessible across different languages and cultures, using only simple audio recordings.
We plan to collect regional voice samples and label them using standard depression scales, which makes our dataset more inclusive.
The entire system can be developed using open-source frameworks like TensorFlow and OpenSMILE.
We only need microphones and regular laptops for processing — so hardware cost is minimal.
The most time-intensive part will be collecting balanced and unbiased voice data.”

⚠️ ChallengesHow to Speak:

“Some of the main challenges we expect are collecting balanced and unbiased voice samples, since accent and tone differ across people.
Another key challenge is distinguishing between short-term sadness and long-term depressive speech patterns.
We’ll also deal with real-world noise in recordings and ensure participant privacy through consent forms and anonymized storage.
Later, when we combine facial features, syncing the audio and video precisely will also be challenging.”

📚 Literature Survey — Study 1How to Speak:

“The first study we looked at is from Carnegie Mellon University by Cohn et al.
They analyzed 57 clinically diagnosed patients using facial expressions and voice tone.
They applied FACS and AAM for facial features, and prosody analysis for speech, classifying results with an SVM.
They achieved around 79–88 percent accuracy, showing that both facial and vocal cues are strong indicators of depression.
However, the limitation was that their dataset was only in English and relatively small.”

📚 Literature Survey — Study 2 How to Speak:

“The second study is from MIT Lincoln Lab.
They used the AVEC 2013 dataset, which includes audio, video, and text, all labeled using the Beck Depression Inventory.
They extracted audio features using OpenSMILE, facial landmarks for visual cues, and linguistic features using LIWC for text.
Their model, based on Support Vector Regression, achieved a correlation of about 0.65 when combining all three inputs — showing that multimodal fusion improves performance.
However, it was still limited to English speakers and lab-based data.”

1. How will you collect your data?
We'll start by using open, research-based datasets like **AVEC** and **DAIC-WOZ** that already contain labeled depression voice samples. Later, we plan to record a small multilingual dataset within our campus — mainly in **English and Kannada** — with proper consent forms to ensure the data collection is ethical and safe.

2. What's new in your project?
Our novelty lies in extending depression detection to **multilingual and regional voices**, which most existing research ignores. We're also planning to add **context-aware modeling** and an element of **explainable AI**, so the model can show which features (like pitch or pauses) influenced the prediction.

3. Isn't this already done?
Yes, similar systems exist, but they're almost always **English-only and conducted in lab environments**. Our focus is to **adapt the model for Indian voices** and make it suitable for **real-world usage**, where background noise and natural speaking variations are common.

4. What algorithms will you use?
We'll implement the system using **TensorFlow or PyTorch** for machine learning. For feature extraction, we'll use **OpenSMILE** to get MFCC, jitter, shimmer, and pitch-related features. For classification, we'll experiment with models like **SVM**, **CNN**, and **LSTM** to identify which performs best.

5. How feasible is data labeling?
We'll rely on standard **clinical depression scales** such as **BDI-II** (Beck Depression Inventory) or **PHQ-9** for labeling data. We'll seek **guidance from mental health professionals** to ensure labeling accuracy and ethical compliance.

6. How do you handle privacy and ethics?
All collected voice data will be taken **with informed consent**, stored **anonymously**, and used **strictly for academic research**. We'll avoid collecting any personally identifiable details and ensure recordings are kept securely and confidentially.

7. Why combine voice and face?
Because **multimodal systems** capture both **verbal (voice)** and **non-verbal (facial expression)** cues. Research shows that combining these two improves detection accuracy by **6–9%**, as they complement each other — sometimes facial emotion is subtle, but voice reveals deeper emotional states.

8. What's your expected accuracy?
Based on previous work from **MIT Lincoln Lab** and the **AVEC challenge**, we expect an accuracy between **70 to 80 percent** in the initial phase, depending on dataset size and noise quality. With multilingual fine-tuning, accuracy may further improve.

9. What challenges are you expecting?
The main challenges include **dataset bias**, **noisy real-world audio**, and differentiating **temporary sadness from actual depression**. We plan to manage these by improving dataset balance, applying denoising filters, and including contextual modeling to identify consistent depressive speech patterns.

10. How is this relevant socially?
Our system can enable **early, stigma-free mental health screening** just through normal voice conversations. It could be integrated into **telemedicine apps, counseling platforms, or educational institutions**, helping people seek help earlier — potentially preventing serious outcomes like self-harm or suicide.